

TARAWA BONRIKI, Kiribati

WMO: 916100

Lat: 1.382N

Lon: 173.147E

Elev: 9

StdP: 14.69

Time Zone: 12.00 (E12)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	75.9	76.8	69.7	109.4	81.3	70.5	112.5	81.7	25.7	83.8	22.7	83.5	6.3	70	0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	8.4	89.9	80.0	89.6	79.9	89.1	79.8	82.4	87.4	81.7	87.0	81.2	86.7	9.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.8	160.2	86.2	80.1	156.3	85.7	79.5	153.0	85.1	46.1	87.7	45.4	87.2	45.0	87.5	89.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(4) 22.9	20.3	18.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			70.1	84.4	2.9	2.1	68.0	85.9	66.3	87.2	64.7	88.3	62.6	89.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	83.5	83.1	83.1	82.8	83.1	83.6	83.6	83.6	83.9	83.8	83.8	83.4	(6)	
(7)		DBStd	1.56	1.55	1.69	1.63	1.53	1.43	1.45	1.43	1.45	1.47	1.42	1.47	1.73	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	12208	1025	926	1017	994	1040	1009	1040	1042	1018	1048	1014	1035	(10)
(11)		CDD65	6733	560	506	552	544	575	559	575	577	568	583	564	570	(11)
(12)		CDH74	79993	6455	5776	6280	6291	6883	6735	7015	7014	6965	7062	6796	6721	(12)
(13)	CDH80	29149	2177	1904	2008	2123	2530	2549	2691	2682	2742	2729	2598	2416	(13)	
(14)	Wind	WSAvg	9.5	11.3	11.9	10.7	9.7	8.9	8.4	8.3	9.6	8.6	9.4	9.3	(14)	
(15)	Precipitation	PrecAvg	67.2	7.5	5.7	7.6	7.2	5.0	4.4	4.6	3.7	3.9	4.4	7.9	(15)	
(16)		PrecMax	128.9	20.1	17.7	21.2	20.4	14.6	12.8	10.0	14.3	12.4	15.5	14.6	20.2	(16)
(17)		PrecMin	20.2	1.2	0.3	0.9	0.7	1.4	1.2	1.3	0.4	0.3	0.2	0.4	0.4	(17)
(18)		PrecStd	33.0	5.6	4.4	5.9	5.4	3.4	3.3	2.4	3.3	3.0	4.0	3.5	6.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.6	90.0	89.7	89.1	89.8	89.9	89.9	90.0	90.0	90.2	90.0	(19)	
(20)			MCWB	80.4	80.2	80.6	80.2	80.2	80.4	80.3	79.7	79.5	79.9	79.8	80.4	(20)
(21)		2%	DB	88.3	88.9	88.2	88.2	88.8	89.0	89.1	89.3	89.5	89.4	89.5	89.2	(21)
(22)			MCWB	79.8	79.8	80.0	80.0	80.1	79.9	80.1	79.6	79.3	79.5	79.5	80.2	(22)
(23)		5%	DB	87.7	87.8	87.3	87.6	88.1	88.2	88.2	88.3	88.5	88.5	88.7	88.2	(23)
(24)			MCWB	79.5	79.4	79.7	79.9	79.9	79.7	79.9	79.3	79.2	79.1	79.3	79.9	(24)
(25)		10%	DB	86.7	86.6	86.3	86.6	87.4	87.7	87.7	87.7	87.8	87.8	87.9	87.4	(25)
(26)			MCWB	79.0	79.0	79.3	79.5	79.8	79.6	79.8	79.1	79.1	78.9	79.1	79.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	82.6	82.3	82.5	82.3	82.5	82.5	82.1	82.1	82.2	81.9	82.3	82.6	(27)
(28)			MADB	87.0	87.2	87.2	87.0	87.3	87.2	87.4	87.1	87.4	87.9	88.1	87.7	(28)
(29)		2%	WB	81.3	81.1	81.3	81.2	81.4	81.3	81.2	80.9	81.0	80.9	81.1	81.6	(29)
(30)			MADB	86.4	86.1	86.2	86.2	86.6	86.7	87.0	87.1	87.2	87.5	87.4	87.1	(30)
(31)		5%	WB	80.6	80.4	80.7	80.7	80.8	80.7	80.7	80.2	80.4	80.1	80.5	80.9	(31)
(32)			MADB	86.0	85.5	85.6	85.9	86.3	86.4	86.8	86.6	86.8	86.7	86.9	86.4	(32)
(33)		10%	WB	79.9	79.8	80.0	80.0	80.2	80.1	80.1	79.7	79.8	79.5	79.8	80.1	(33)
(34)			MADB	85.2	84.9	84.9	85.2	85.6	85.7	86.0	86.0	86.1	86.0	85.9	85.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.8	7.5	7.7	7.6	7.9	8.4	8.4	8.4	8.7	8.4	8.1	(35)	
(36)			MCDBR	8.3	8.3	8.4	8.1	8.6	9.0	9.1	8.9	9.0	9.3	9.2	8.8	(36)
(37)			MCWBR	4.7	4.3	4.5	3.9	4.1	4.3	4.4	4.4	4.3	4.5	4.4	4.4	(37)
(38)		5% WB	MCDBR	8.1	7.8	8.0	7.8	8.2	8.8	8.7	8.7	8.6	9.0	8.8	8.5	(38)
(39)			MCWBR	5.0	4.6	4.9	4.2	4.3	4.6	4.5	4.5	4.6	4.8	4.5	4.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.423	0.424	0.424	0.403	0.389	0.384	0.384	0.380	0.384	0.379	0.389	0.416	(40)	
(41)		taud	2.499	2.487	2.514	2.586	2.627	2.640	2.642	2.646	2.617	2.631	2.602	2.524	(41)	
(42)		Ebn at Noon	287	288	286	286	283	281	282	289	293	298	296	288	(42)	
(43)		Edh at Noon	36	36	36	32	30	29	29	30	31	31	32	34	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1754	1877	1852	1886	1885	1809	1821	1946	2044	2050	1951	1739	(44)	
(45)		RadStd	266	243	288	195	123	139	128	148	151	166	137	257	45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page